

Weather Trend Forecasting Assessment

Final Report

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Completed: Basic and Advanced Assessment

1. Project Summary

This project analyzes the Global Weather Repository dataset from Kaggle to explore global weather patterns and forecast daily average temperature trends. I built the project as a complete data science workflow, starting from data cleaning and exploratory analysis, then moving into time-series forecasting and advanced analysis.

The main forecasting target is the daily average temperature in Celsius. I used the `last\_updated` column as the time-series field, created lag features and rolling averages, compared multiple models, and evaluated the results with MAE, RMSE, MAPE, and R2.

The best-performing model in my run was Linear Regression, with MAE = 0.225, RMSE = 0.668, MAPE = 1.95%, and R2 = 0.938. The Ensemble model also performed well, but it did not outperform Linear Regression. This showed that strong feature engineering can allow a simpler model to perform better than more complex models.

2. PM Accelerator Mission

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3. Dataset and Objective

Dataset: Global Weather Repository / World Weather Repository from Kaggle.

Objective: Analyze global daily weather data, uncover weather trends and patterns, forecast future temperature trends, and demonstrate both basic and advanced data science techniques.

- Basic tasks completed: cleaning, missing-value handling, outlier handling, preprocessing, EDA, visualizations, and forecasting.
- Advanced tasks completed: anomaly detection, multiple forecasting models, ensemble model, climate analysis, air-quality analysis, feature importance, and geographical analysis.
- Main time-series column: last_updated.
- Main forecasting target: temperature_celsius.

4. Project Workflow and File Roles

The project is organized so that each file has a clear responsibility.

File or Folder	Purpose
main.py	Runs the complete end-to-end pipeline.
app.py	Creates the Streamlit dashboard with project sections, charts, metrics, and final insights.
src/data_cleaning.py	Loads the dataset, standardizes column names, handles missing values, removes duplicates, converts last_updated to datetime, and caps outliers.
src/eda.py	Creates exploratory charts for temperature, precipitation, correlations, and country-level distributions.
src/forecasting.py	Creates time-series features, trains forecasting models, evaluates metrics, and saves forecast plots and feature importance.
src/advanced_analysis.py	Runs anomaly detection, climate analysis, air-quality correlation analysis, and geographical analysis.
outputs/	Stores generated CSV summaries and chart images.
README.md	Documents the project, setup instructions, methodology, results, and how to reproduce the work.

5. Data Cleaning and Preprocessing

I cleaned the dataset before modeling because weather data can contain inconsistent column names, missing values, duplicates, invalid dates, and extreme numeric values. Cleaning was important to make the downstream EDA and forecasting steps more reliable.

- Standardized column names by converting them to lowercase and replacing spaces or hyphens with underscores.
- Converted last_updated into a datetime column so it could be used for time-series analysis.
- Removed duplicate records.
- Filled missing numeric values with the median value of each column.
- Filled missing categorical values with Unknown.
- Removed rows with invalid or missing dates.
- Capped numeric outliers using percentile-based clipping.
- Saved the cleaned dataset and cleaning summary to the outputs folder.

Report image note: no chart is required for this section unless a missing-values chart is generated. If available, insert outputs/figures/missing_values.png directly after this paragraph.

6. Exploratory Data Analysis

The EDA step helped me understand trends, correlations, and patterns before building forecasting models. I focused on temperature, precipitation, country-level record distribution, and relationships between numeric weather variables.

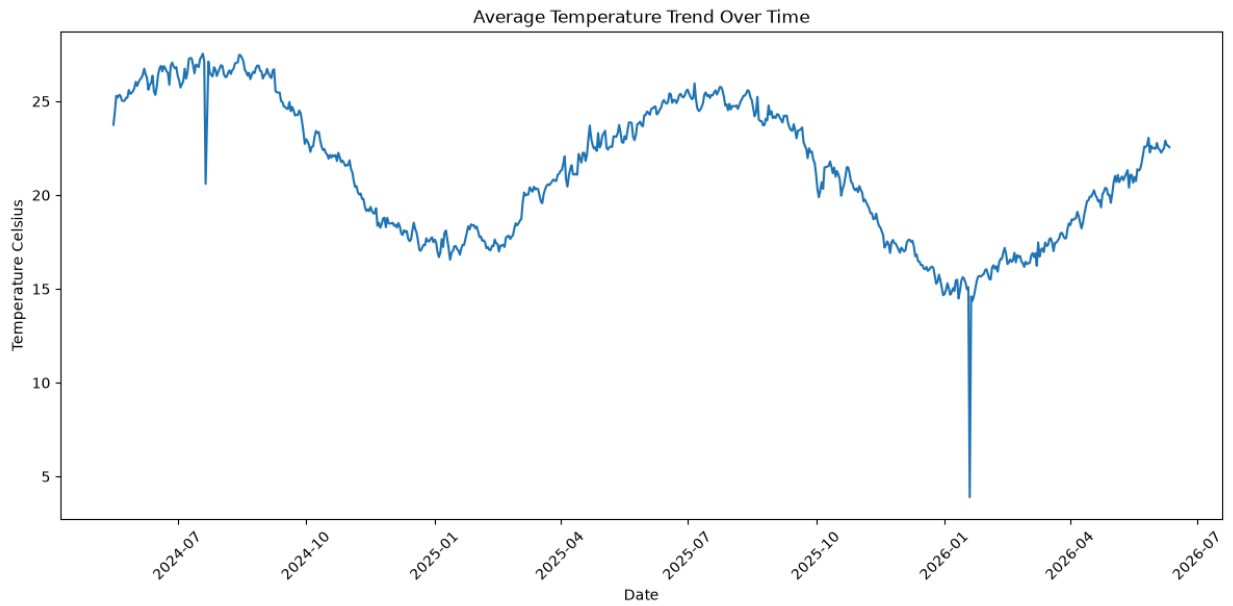


Figure 1. Average Temperature Trend Over Time.

This chart shows the daily average temperature pattern across the dataset. The visible seasonal movement supports using time-based features such as month, lag values, and rolling averages.

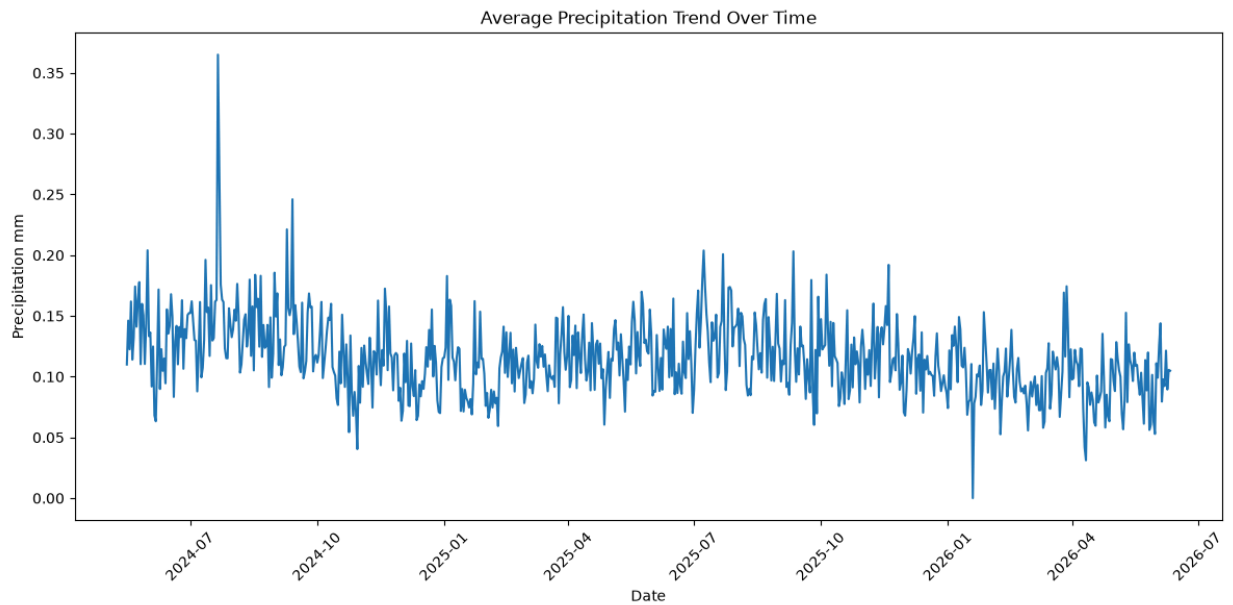


Figure 2. Average Precipitation Trend Over Time.

This chart shows precipitation changes over time. Compared with temperature, precipitation is more irregular and includes short spikes.

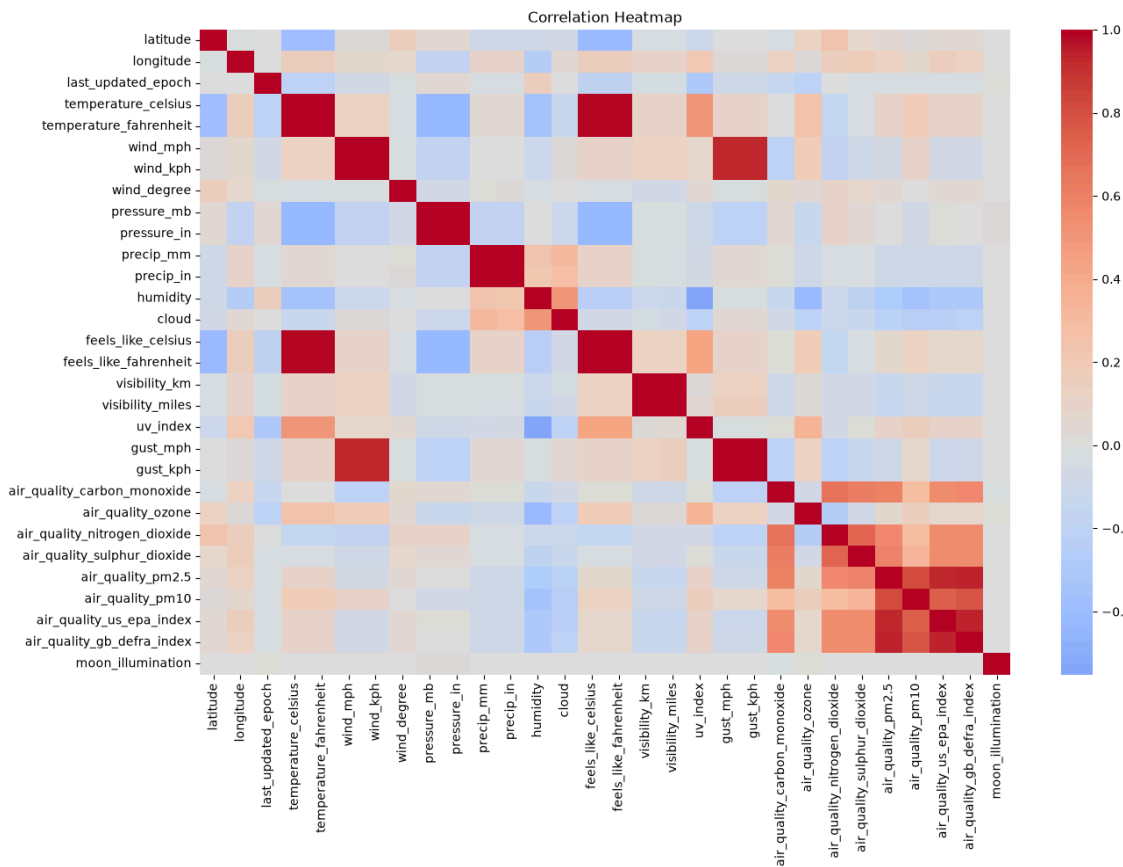


Figure 3. Correlation Heatmap.

This heatmap shows relationships between numeric weather and air-quality variables. Strong relationships appear among related units and derived weather variables, while other variables show weaker relationships.

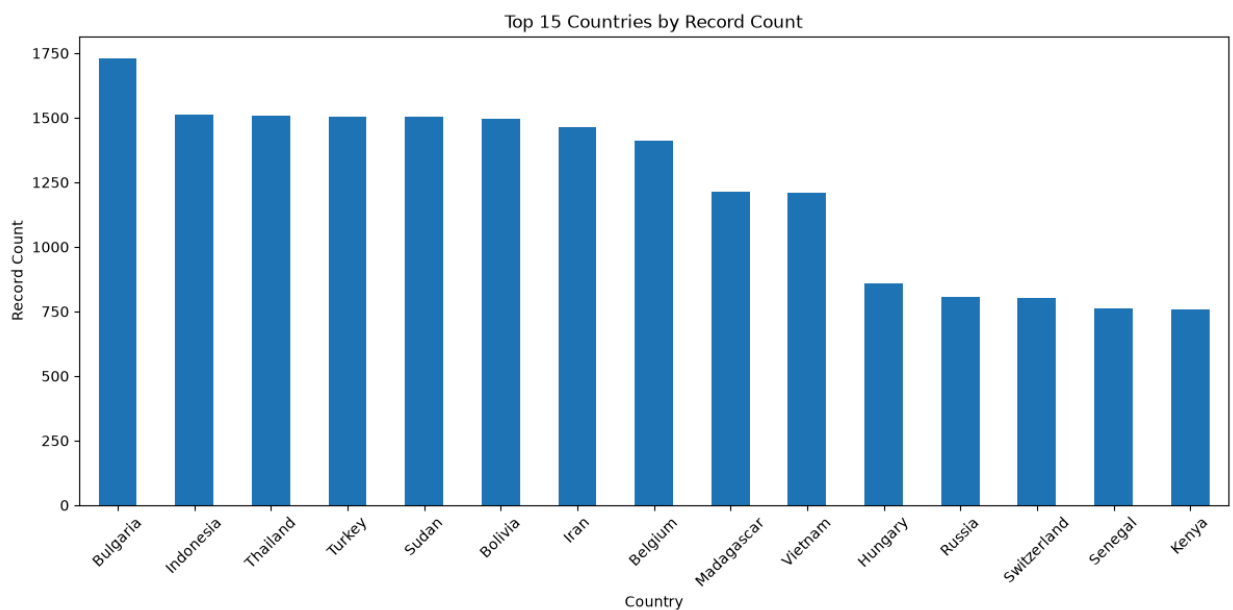


Figure 4. Top 15 Countries by Record Count.

This chart shows which countries have the most records in the dataset. Reviewing record distribution is important because uneven coverage can influence global averages.

7. Forecasting Methodology

For forecasting, I used `last_updated` as the time-series column and `temperature_celsius` as the main target. Because the dataset contains many cities per date, I aggregated the records into daily average temperature before modeling.

I created lag features and rolling-average features because recent weather is usually a strong predictor of near-term weather. I also added calendar features such as day of week and month to capture time-based patterns.

- `lag_1`: previous day temperature.
- `lag_3`: temperature from three days earlier.
- `lag_7`: temperature from seven days earlier.
- `rolling_3`: average temperature over the recent three-day window.
- `rolling_7`: average temperature over the recent seven-day window.
- `day_of_week` and `month`: calendar-based features.

I used a chronological train/test split rather than a random split. This matters because in time-series forecasting, the model should train on older data and be evaluated on newer data to avoid future data leakage.

8. Models Compared and Evaluation Results

I compared simple baselines with machine learning models. The baselines helped me check whether the more advanced models were actually improving the forecast.

Model	MAE	RMSE	MAPE	R2
Naive Baseline	0.393	1.304	3.75%	0.764
Moving Average	0.400	0.932	3.55%	0.879
Linear Regression	0.225	0.668	1.95%	0.938
Random Forest	0.301	0.939	3.15%	0.877
Gradient Boosting	0.289	0.936	3.06%	0.878
Ensemble	0.253	0.776	2.61%	0.916

Linear Regression was the best-performing model, with the lowest RMSE and highest R2. This result was useful because it showed that the engineered lag, rolling-average, and calendar features captured much of the daily temperature structure. The Ensemble model was strong, but it did not beat Linear Regression.

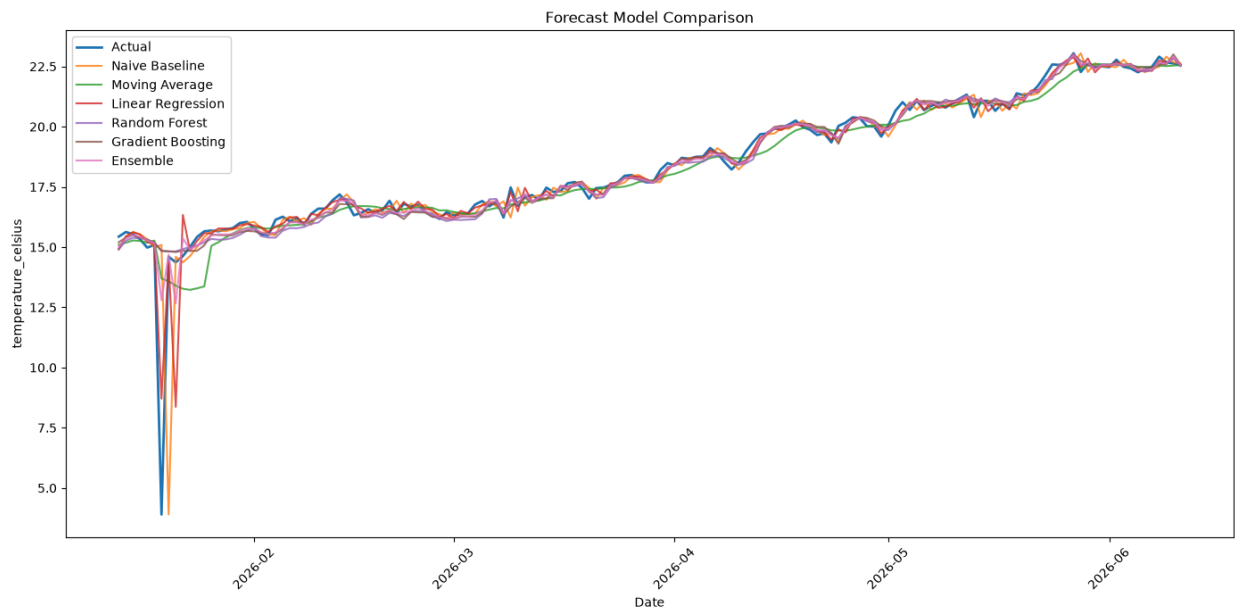


Figure 5. Forecast Model Comparison.

This figure compares the actual daily average temperature with predictions from the baseline models, machine learning models, and ensemble model. Most models follow the trend closely, but Linear Regression produced the best overall metric performance.

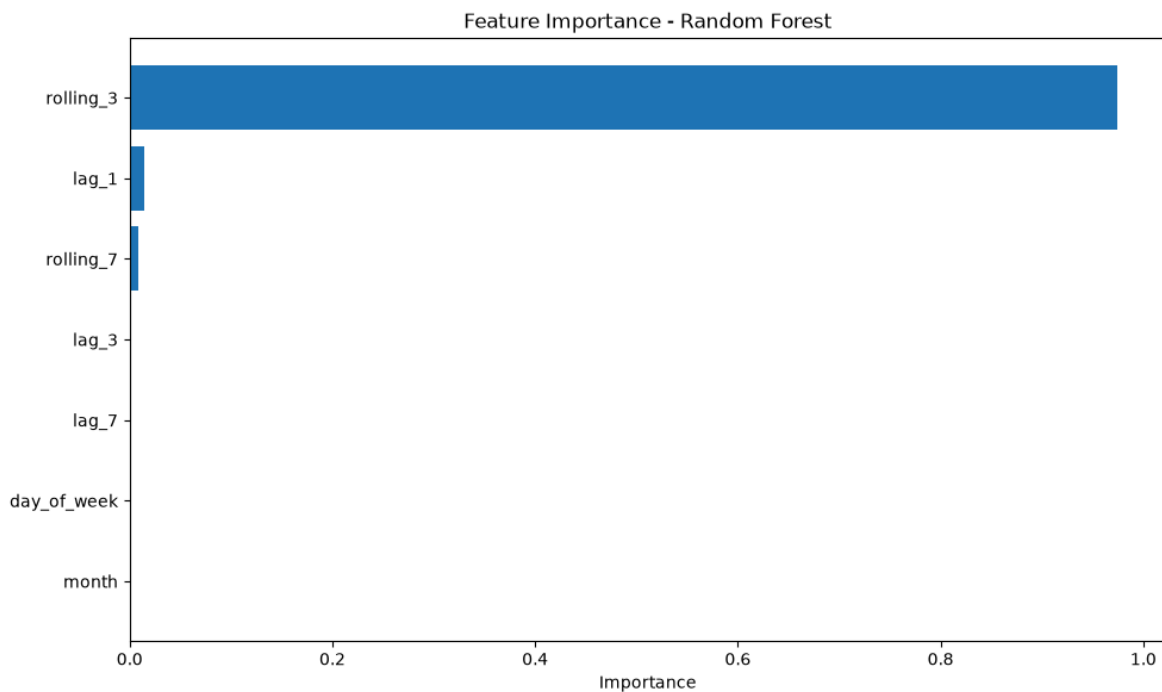


Figure 6. Feature Importance - Random Forest.

The Random Forest feature importance plot shows that rolling_3 was the strongest feature. This supports the idea that recent average temperature is highly useful for temperature forecasting.

9. Advanced Analysis

9.1 Anomaly Detection

I used anomaly detection to identify unusual weather records. These records may represent extreme weather events, unusual combinations of weather variables, or possible data quality issues.

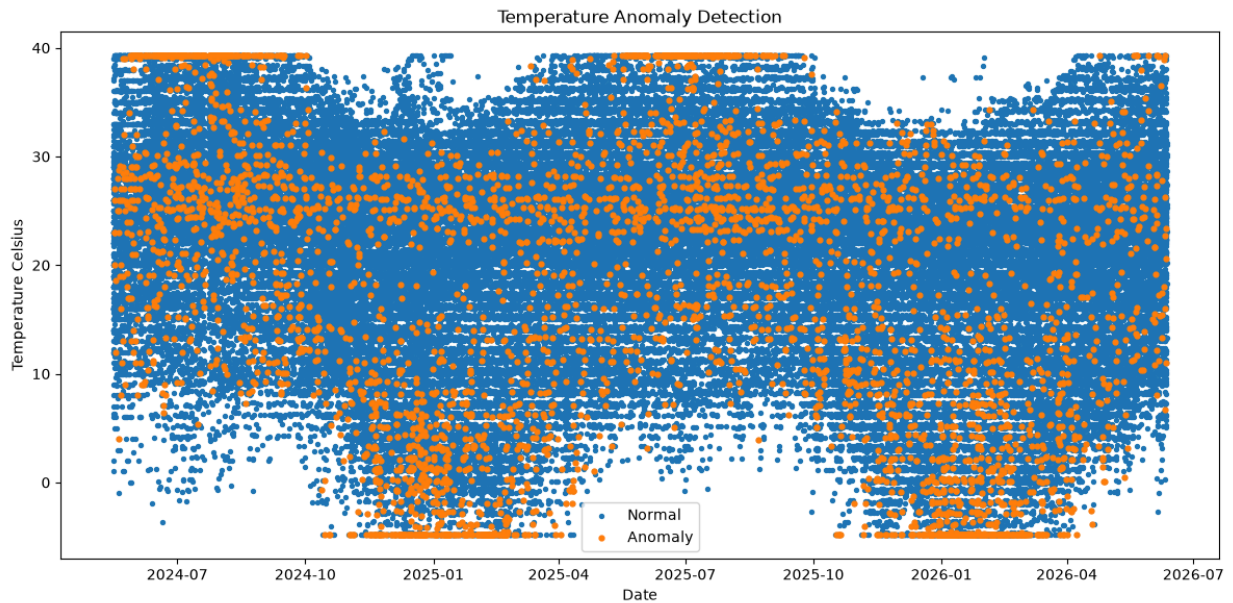


Figure 7. Temperature Anomaly Detection.

The anomaly plot highlights unusual temperature observations over time. These points are useful to review because extreme or unusual values can influence both EDA and model performance.

9.2 Climate Analysis

For climate analysis, I grouped the data by month and country to study broader patterns rather than only daily changes. This helped compare average temperature, precipitation, humidity, and wind speed across time periods and locations.

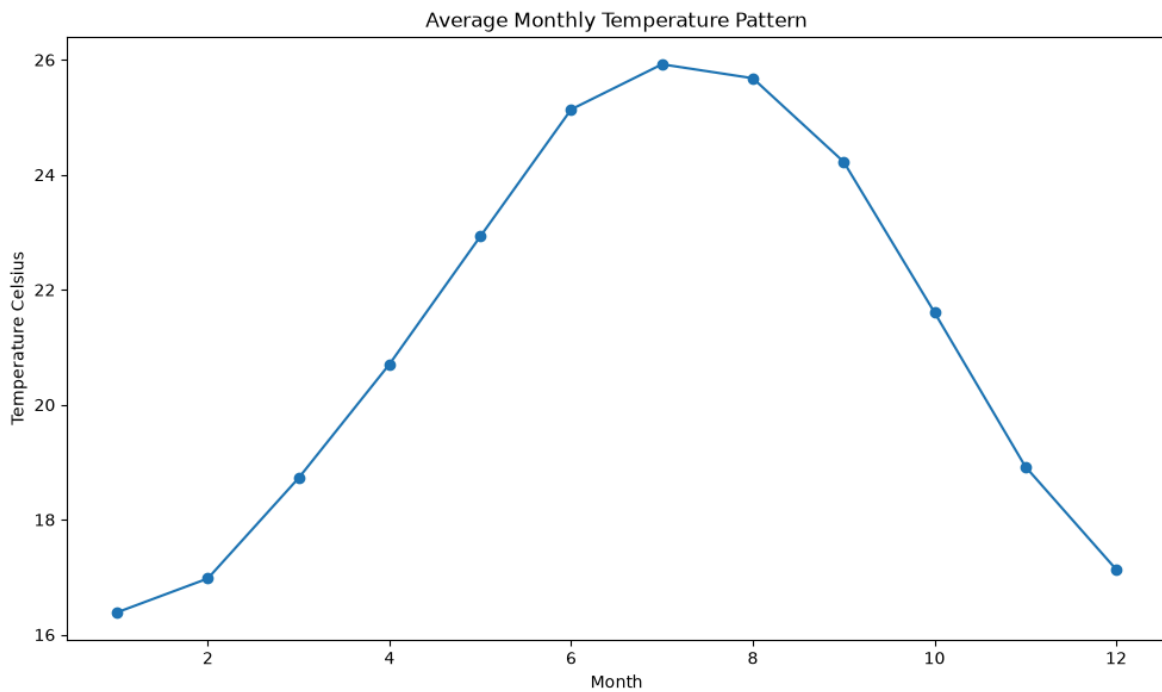


Figure 8. Average Monthly Temperature Pattern.

The monthly temperature pattern shows a clear rise toward the middle of the year and a decline toward the end of the year, which supports the presence of seasonal structure in the data.

9.3 Air-Quality and Environmental Analysis

For air-quality analysis, I calculated correlations between weather variables and air-quality indicators. I used this as an exploratory step to see whether pollution-related variables moved with temperature, humidity, wind speed, pressure, or precipitation.

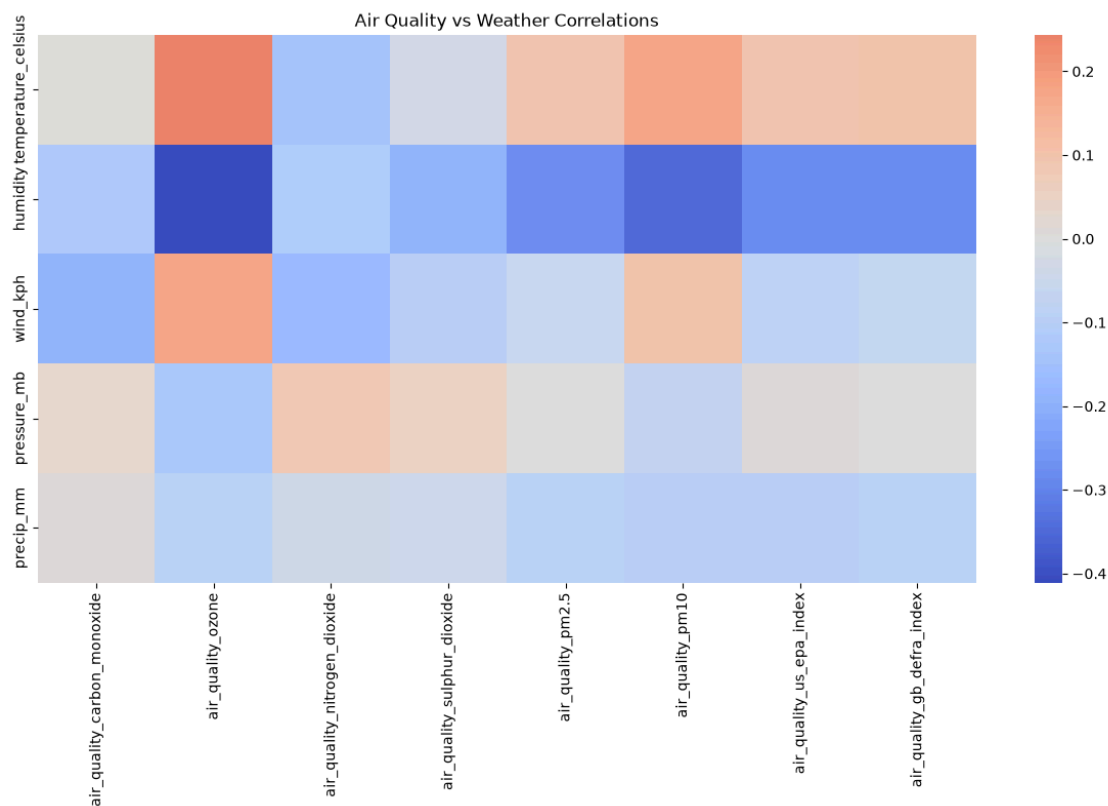


Figure 9. Air Quality vs Weather Correlations.

This heatmap shows relationships between weather variables and air-quality indicators. The analysis is correlational, so it does not prove causation, but it gives useful direction for environmental interpretation.

9.4 Geographical Pattern Analysis

For geographical analysis, I grouped the dataset by country to compare weather patterns across locations. This helped show that global averages can hide important country-level differences.

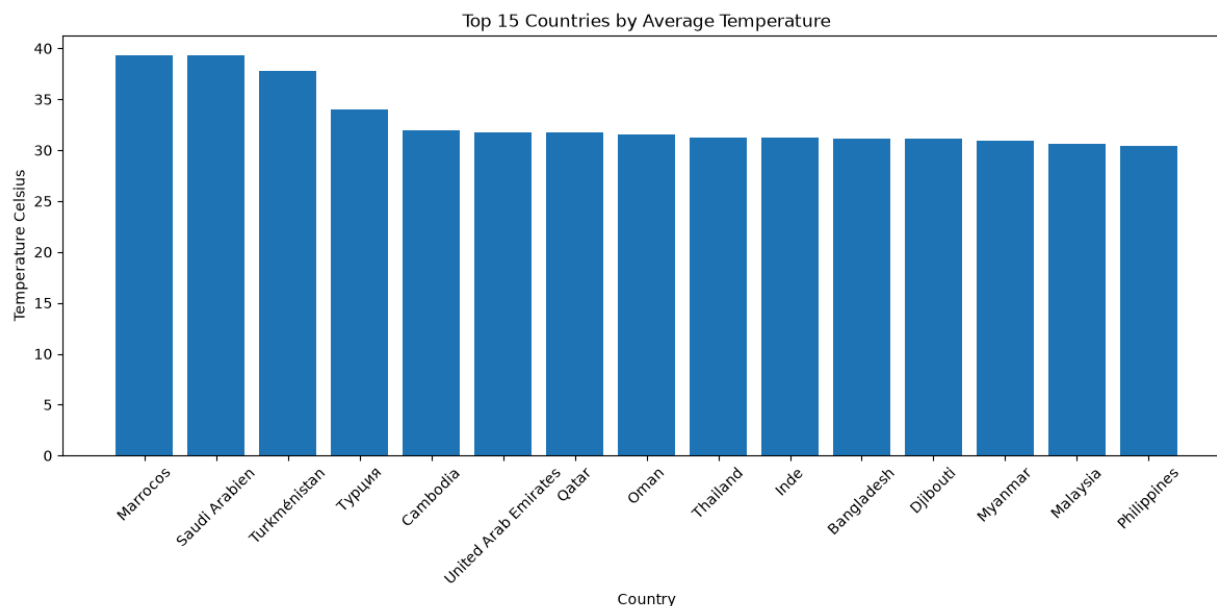


Figure 10. Top 15 Countries by Average Temperature.

This chart shows the countries with the highest average temperatures in the dataset. Morocco, Saudi Arabia, and Turkmenistan appear among the warmest countries in this analysis.

10. Key Insights

- Linear Regression was the best model in this run, with RMSE = 0.668 and R2 = 0.938.
- The Ensemble model performed well, but it did not outperform Linear Regression.
- The Moving Average model also performed strongly, which makes sense because temperature usually changes gradually.
- The feature importance analysis showed rolling_3 as the strongest Random Forest feature, suggesting recent average temperature is highly useful for forecasting.
- The temperature trend and monthly climate chart showed clear seasonal movement.
- Precipitation was more irregular than temperature and included short spikes.
- Air-quality correlation analysis provided exploratory environmental context but should not be interpreted as causation.
- Geographical analysis showed that weather patterns differ clearly across countries.

11. Limitations

- The main forecast uses global daily average temperature, so it does not capture detailed city-level forecasting.
- The dataset combines many countries and cities, which can smooth out local patterns.

- The model does not include external climate factors such as elevation, ocean patterns, or seasonal climate indexes.
- Correlation analysis does not prove causation.
- The forecasting models are simple and interpretable; more advanced time-series models could be tested in future work.

12. Future Improvements

- Forecast temperature for individual cities or countries.
- Test additional time-series methods such as ARIMA, SARIMA, or Prophet.
- Add more detailed interactive maps for spatial analysis.
- Improve feature engineering with location-based variables.
- Add model saving and future prediction scripts.
- Compare model performance across different countries or climate regions.

13. Reproducibility

1. Clone the GitHub repository.
2. Create and activate a Python virtual environment.
3. Install dependencies using `pip install -r requirements.txt`.
4. Download the Kaggle dataset and place it at `data/GlobalWeatherRepository.csv`.
5. Run `python main.py --data data/GlobalWeatherRepository.csv`.
6. Launch the dashboard with `streamlit run app.py`.